Learning Outcome of the Assignment

1. Compare various techniques used in the defense of computer systems against malicious software and software-based attacks. (A4, PLO9)

Instructions:

No marks will be awarded for the entire assignment if any part of it is found to be copied directly from printed materials or from another student. All submissions should be made on or before the due date. Any late submissions after the deadline will not be entertained. **Zero (0)** mark will be awarded for late submission unless extenuating circumstances are upheld.

Questions:
You are required to conduct research individually.

Contemporary threat actors are actively abusing the vulnerabilities of Windows OS and sophisticated evasion strategies to avoid antivirus (AV) and Endpoint Detection and Response (EDR) software. Such attacks are often based on kernel exploits, firmware attacks, Living-off-the-Land (LOLbins), vulnerable drivers, fileless malware, and stealth persistence mechanisms that are very hard to detect.

You must perform research on the most recent windows OS exploitation methods (2023 - 2025) which are challenging to identify using antivirus and EDR software.

1. Your research should focus on one of the following areas (but not limited to):
    - BlackLotus UEFI Bootkit — Secure Boot bypass
    - UEFI Secure Boot Bypass Exploits
    - HybridPetya Boot-Level Persistence
    - Firmware Rootkits on Windows 11
    - Early Boot Chain Injection
    - BYOVD — Bring Your Own Vulnerable Driver (RTCore64, TPwSav.sys, Baidu drivers)
    - Signed Driver Abuse for Kernel R/W
    - Kernel Rootkits with Expired Legacy Certificates (ValleyRAT)
    - Kernel Callback Hook Removal
    - Kernel Object Manipulation (EPROCESS hiding)
    - Kernel Memory Patch Attacks
    - Process Protection Bypass via Vulnerable Drivers
    - OneDrive / Teams as Malware C2 Channel
    - Group Policy Object (GPO) Abuse for Lateral Movement
    - Safe Mode Boot Abuse to Disable Security

2. Research on the vulnerability of the Operating System that enables the chosen attacks. Support your research with Mitre Attack framework or Common Vulnerabilities and Exposures (CVE) database.
3. Discuss on Tactics, Techniques and Procedures used by attackers in performing the chosen attacks.
4. Integrating **Operational Security (OPSEC), General and Technical Security, and Machine Learning-based detection** significantly enhances an organization's ability to detect and mitigate **the above exploitation**. As a cyber security expert/consultant provide the following plan for the organization:
    - Operational Security (OPSEC)
    - General and Technical Security
    - Machine learning-based detection or deep learning (compare at least 3 algorithms and identify the best to be implemented)
    - Include diagrams or charts where applicable to illustrate your threat model

Assesment

This assignment will contribute 30% towards the module marks, as mentioned in the Test Specification Table. This assignment will be evaluated based on the following criteria:

Assessment Criteria (Marks Breakdown)

Marking Criteria	Weighting	Marks
Section B	100	
System Vulnerabilities	20	
Tactic, techniques and Procedures implemented by attackers	20	
Operational Security (OPSEC),	20	
Cyber Security (General and Technical Security) and machine learning-based detection	20	
Threat Modelling	20	
Total	100	

Guidelines for the Report:

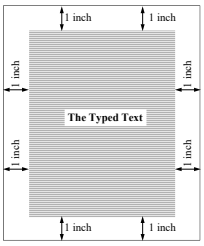
Document the results of your work in a **professional and systematic** manner, in the form of a **computerized report**. **One (1**) softcopy of your documentation is to be submitted.

Your completed documentation should meet the following requirements:

1. Table of contents for every detailed chapter/section.
2. Abstract
3. Introduction
4. Chapters / sections
5. Conclusion
6. Appendices
7. References

Submission requirements

1. Your report must be typed using Microsoft Word with Times New Roman font. You need use to include a word count at the end of the report (excluding title, source code of program & contents pages) Report should be in 1.5 spaces.
2. The report should have a one (1") margin all around the page as illustrated below:



3. Every report must have a *front cover*. The front cover should have the following details:-
   a) Name
   b) Intake code.
   c) Subject.
   d) Project Title.
   e) Date Assigned (the date the report was handed out).
   f) Date Completed (the date the report is due to be handed in).
4. **All** information, figures and diagrams obtained from external sources **must** be referenced using the APA referencing system accordingly.
5. Assignment submission is online submission through Moodle

Marking Scheme Rubrics

Marking Criteria	Fail 0-4 Marks	Pass 5-9 Marks	Credit 10-14 Marks	Distinction 15-20 Marks
Report: Cyber Vulnerabilities (20)	All submission requirements were not adhered or poor writing or poor quality of contents on vulnerabilities. No integration of the tasks given	All requirements are fulfilled but with some missing parts. Research not in detail on vulnerabilities	All requirements are fulfilled. No missing parts. Acceptable research but research lack of supporting details on vulnerabilities	All requirements are fulfilled. No missing parts. Outstanding research with good supporting details on vulnerabilities
Report: Attacks Tactic, Techniques and Procedures (TTPs) (20)	All submission requirements were not adhered or poor writing or poor quality of contents on TTPs. No integration of the tasks given	All requirements are fulfilled but with some missing parts. Research not in detail on TTPs.	All requirements are fulfilled. No missing parts. Acceptable research but research lack of supporting details on TTPs.	All requirements are fulfilled. No missing parts. Outstanding research with good supporting details on TTPs.
Report: Operational Security (OPSEC) (20)	Poor research and investigation of the operational security. Poor evaluation of the requirement.	Acceptable research and investigation are done. Acceptable evaluation of the operational security with proper reasoning with proper planning and management.	Good analysis and investigation are done. Good evaluation of the operational security with proper reasoning. Good planning and management with detail research	Outstanding analysis and investigation of the problem. Outstanding evaluation of the operational security with proper reasoning. Outstanding planning and management with detail research
Report: Cyber Security (General and Technical Security) (20)	Poor research and investigation of the General and Technical Security security. Poor evaluation of the requirement.	Acceptable research and investigation are done. Acceptable evaluation of the General and Technical Security with proper reasoning	Good analysis and investigation are done. Good evaluation of the General and Technical Security with proper reasoning. Good	Outstanding analysis and investigation of the problem. Outstanding evaluation of the General and Technical Security with

		with proper planning and management.	planning and management with detail research	proper reasoning. Outstanding planning and management with detail research
Presentation: Threat Modeling (20)	Poor content of research on Threat Modelling. Demonstrate dependency on others guidance during presentation. Unable to answer any questions independently.	Acceptable content of research on Threat Modelling. Demonstrate tendency to dependent on others guidance during presentation. Able to answer question but failed to produce confirmed answers	Good content of research on Threat Modelling. Demonstrate good understanding and idea in the topic area. Able to answer the questions but answer given lack of supporting details.	Outstanding content of research on Threat Modelling. Always demonstrate a self-reliant attitude in all situations during presentation. Voice is clear and loud. Able to answer all the questions without referring to notes.